

Posterior Derivation Notes  
Bayesian Hierarchical Logistic Regression  
for 30-Day Hospital Readmission

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April 2026

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# 1 Notation

| Symbol                                | Meaning                                        |
|---------------------------------------|------------------------------------------------|
| $N = 37,964$                          | Number of observations (encounters)            |
| $K = 43$                              | Number of patient-level covariates             |
| $J = 19$                              | Number of medical specialty groups             |
| $y_i \in \{0, 1\}$                    | 30-day readmission indicator for encounter $i$ |
| $\mathbf{x}_i \in \mathbb{R}^K$       | Covariate vector for encounter $i$             |
| $j[i] \in \{1, \dots, J\}$            | Specialty group index for encounter $i$        |
| $n_j$                                 | Number of observations in group $j$            |
| $\alpha \in \mathbb{R}$               | Global intercept                               |
| $\boldsymbol{\beta} \in \mathbb{R}^K$ | Fixed-effect coefficients                      |
| $u_j \in \mathbb{R}$                  | Random intercept for specialty group $j$       |
| $\mathbf{u} = (u_1, \dots, u_J)^\top$ | Vector of all random intercepts                |
| $\tau > 0$                            | Between-group standard deviation               |
| $\eta_i$                              | Linear predictor for observation $i$           |

## 2 Model Specification

### 2.1 Likelihood (Data Model)

Conditional on the parameters, outcomes are independent Bernoulli:

$$y_i \mid \alpha, \boldsymbol{\beta}, \mathbf{u} \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\pi_i), \quad i = 1, \dots, N, \quad (1)$$

where the success probability is linked to the linear predictor through the logistic function:

$$\eta_i = \alpha + \mathbf{x}_i^\top \boldsymbol{\beta} + u_{j[i]}, \quad \pi_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}. \quad (2)$$

The full data likelihood is therefore:

$$p(\mathbf{y} \mid \alpha, \boldsymbol{\beta}, \mathbf{u}) = \prod_{i=1}^N \pi_i^{y_i} (1 - \pi_i)^{1-y_i} = \prod_{i=1}^N \frac{\exp(y_i \eta_i)}{1 + \exp(\eta_i)}. \quad (3)$$

### 2.2 Prior Distributions

We place weakly informative, independent priors on each parameter block.

**Intercept.**

$$\alpha \sim \mathcal{N}(0, \sigma_\alpha^2), \quad \sigma_\alpha = 5. \quad (4)$$

**Fixed effects.**

$$\beta_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, s_\beta^2), \quad k = 1, \dots, K, \quad s_\beta = 2.5. \quad (5)$$

**Random intercepts (conditional on  $\tau$ ).**

$$u_j \mid \tau \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \tau^2), \quad j = 1, \dots, J. \quad (6)$$

This is the core hierarchical component: the  $u_j$  are exchangeable given the shared scale  $\tau$ , which induces partial pooling of the specialty effects toward zero.

**Between-group scale (hyperprior).** The baseline prior is

$$\tau \sim \text{Half-Normal}(0, s_\tau^2), \quad s_\tau = 1.0, \quad (7)$$

with density  $p(\tau) = \frac{2}{\sqrt{2\pi} s_\tau} \exp(-\frac{\tau^2}{2s_\tau^2})$  for  $\tau > 0$ .

As a sensitivity alternative (Section 10), we also consider

$$\tau \sim \text{Half-Cauchy}(0, s_\tau), \quad p(\tau) = \frac{2}{\pi s_\tau} \frac{1}{1 + \tau^2/s_\tau^2}, \quad \tau > 0. \quad (8)$$

## 2.3 Assumptions

- (i) Conditional independence:  $y_i \perp y_{i'} \mid \alpha, \beta, \mathbf{u}$  for  $i \neq i'$ .
- (ii) Each patient appears only once in the cohort (first encounter per patient retained).
- (iii) Specialty group assignment  $j[i]$  is treated as fixed (not modelled).
- (iv) The random intercepts  $u_j$  capture unobserved specialty-level heterogeneity; no random slopes are included.
- (v) Prior independence:  $\alpha$ ,  $\beta$ , and  $\tau$  are mutually independent *a priori*;  $\mathbf{u}$  depends on  $\tau$  only.

## 2.4 Hyperparameters

| Symbol          | Value | Role                                                                   |
|-----------------|-------|------------------------------------------------------------------------|
| $\sigma_\alpha$ | 5.0   | Prior SD for the intercept; covers plausible baseline log-odds         |
| $s_\beta$       | 2.5   | Prior SD for each fixed effect; allows OR up to $\sim e^5 \approx 150$ |
| $s_\tau$        | 1.0   | Scale of the half-normal prior on $\tau$ ; concentrates mass near zero |

Table 1: Hyperparameter settings and their interpretation.

# 3 Joint Distribution and Log-Posterior

## 3.1 Joint Distribution

By the conditional independence structure of the model:

$$p(\mathbf{y}, \alpha, \beta, \mathbf{u}, \tau \mid \mathbf{X}) = \underbrace{p(\mathbf{y} \mid \alpha, \beta, \mathbf{u}, \mathbf{X})}_{\text{likelihood}} \underbrace{p(\alpha)}_{\text{prior on } \alpha} \underbrace{p(\beta)}_{\text{prior on } \beta} \underbrace{p(\mathbf{u} \mid \tau)}_{\text{random effects}} \underbrace{p(\tau)}_{\text{hyperprior}}. \quad (9)$$

### 3.2 Posterior Distribution (Up to Proportionality)

By Bayes' theorem:

$$p(\alpha, \beta, \mathbf{u}, \tau \mid \mathbf{y}, \mathbf{X}) \propto p(\mathbf{y} \mid \alpha, \beta, \mathbf{u}, \mathbf{X}) p(\alpha) p(\beta) p(\mathbf{u} \mid \tau) p(\tau). \quad (10)$$

### 3.3 Log-Posterior

Taking logarithms and substituting Eqs. (3)–(7):

$$\begin{aligned} \ell(\alpha, \beta, \mathbf{u}, \tau) &= \log p(\alpha, \beta, \mathbf{u}, \tau \mid \mathbf{y}, \mathbf{X}) + \text{const.} \\ &= \underbrace{\sum_{i=1}^N [y_i \eta_i - \log(1 + e^{\eta_i})]}_{\text{(I) log-likelihood}} - \underbrace{\frac{\alpha^2}{2\sigma_\alpha^2}}_{\text{(II) prior on } \alpha} - \underbrace{\frac{1}{2s_\beta^2} \sum_{k=1}^K \beta_k^2}_{\text{(III) prior on } \beta} \\ &\quad - \underbrace{J \log \tau + \frac{1}{2\tau^2} \sum_{j=1}^J u_j^2}_{\text{(IV) } p(\mathbf{u} \mid \tau)} - \underbrace{\frac{\tau^2}{2s_\tau^2}}_{\text{(V) hyperprior on } \tau}. \end{aligned} \quad (11)$$

**Remark 1.** The log-likelihood term uses the identity  $\log(\pi_i^{y_i} (1 - \pi_i)^{1-y_i}) = y_i \eta_i - \log(1 + e^{\eta_i})$ , which avoids computing  $\pi_i$  explicitly and is numerically stable when implemented as  $\log(1 + e^{\eta_i}) = \text{logaddexp}(0, \eta_i)$ .

## 4 Metropolis-within-Gibbs Sampling Strategy

The posterior (10) does not have a standard closed form due to the logistic likelihood (non-conjugacy). We use a **Metropolis-within-Gibbs (MwG)** sampler that cycles through five parameter blocks at each iteration, updating each block conditional on the current values of all others.

### 4.1 General Metropolis–Hastings Framework

For a generic block parameter  $\theta$  with full conditional target  $h(\theta) = \log p(\theta \mid \text{rest})$ :

1. Propose  $\theta^* \sim q(\theta^* \mid \theta^{(t)})$ .
2. Compute  $\log R = h(\theta^*) - h(\theta^{(t)}) + \log q(\theta^{(t)} \mid \theta^*) - \log q(\theta^* \mid \theta^{(t)})$ .
3. Set  $\theta^{(t+1)} = \theta^*$  with probability  $\min(1, e^{\log R})$ ; otherwise keep  $\theta^{(t)}$ .

All our proposals use **symmetric random walks**  $\theta^* = \theta^{(t)} + \epsilon$ ,  $\epsilon \sim \mathcal{N}(0, \sigma_{\text{prop}}^2)$ , so  $q(\theta^* \mid \theta) = q(\theta \mid \theta^*)$  and the Hastings ratio cancels:

$$\log R = h(\theta^*) - h(\theta^{(t)}). \quad (12)$$

### 4.2 Block 1: Update $\alpha$ (Intercept)

**Full conditional.** Collecting all terms in (11) that depend on  $\alpha$ :

$$h(\alpha) = \sum_{i=1}^N [y_i \eta_i - \log(1 + e^{\eta_i})] - \frac{\alpha^2}{2\sigma_\alpha^2}, \quad (13)$$

where  $\eta_i = \alpha + \mathbf{x}_i^\top \beta + u_{j[i]}$ .

**Proposal and acceptance.**

$$\alpha^* = \alpha^{(t)} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_{\alpha, \text{prop}}^2), \quad (14)$$

$$\log R = h(\alpha^*) - h(\alpha^{(t)}). \quad (15)$$

**Computational note.** When evaluating  $h(\alpha^*)$ , we do *not* recompute  $\eta$  from scratch. Instead,  $\eta^* = \eta^{(t)} + (\alpha^* - \alpha^{(t)})$  shifts every entry by a constant, costing  $O(N)$  rather than  $O(NK)$ .

### 4.3 Block 2: Update $\beta_k$ (Fixed Effects, Component-wise)

Each  $\beta_k$  is updated individually while holding all other components fixed (a *scalar-at-a-time* Gibbs scan within the  $\beta$  block).

**Full conditional for  $\beta_k$ .**

$$h(\beta_k) = \sum_{i=1}^N [y_i \eta_i - \log(1 + e^{\eta_i})] - \frac{\beta_k^2}{2s_\beta^2}. \quad (16)$$

**Proposal and acceptance.**

$$\beta_k^* = \beta_k^{(t)} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_{\beta_k, \text{prop}}^2), \quad (17)$$

$$\log R = h(\beta_k^*) - h(\beta_k^{(t)}). \quad (18)$$

**Computational note.** The linear predictor update is  $\eta_i^* = \eta_i^{(t)} + (\beta_k^* - \beta_k^{(t)}) X_{ik}$ , which costs  $O(N)$  per coordinate rather than  $O(NK)$  for a full recomputation.

### 4.4 Block 3: Update $u_j$ (Random Intercepts, Component-wise)

**Full conditional for  $u_j$ .** Only observations in group  $j$  contribute to the likelihood:

$$h(u_j) = \sum_{i: j[i]=j} [y_i \eta_i - \log(1 + e^{\eta_i})] - \frac{u_j^2}{2\tau^2}. \quad (19)$$

**Proposal and acceptance.**

$$u_j^* = u_j^{(t)} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_{u_j, \text{prop}}^2), \quad (20)$$

$$\log R = h(u_j^*) - h(u_j^{(t)}). \quad (21)$$

**Computational advantage.** The sum in (19) runs over  $n_j$  observations (the size of specialty group  $j$ ), not all  $N$ . For each group we pre-index the relevant observations, reducing cost from  $O(N)$  to  $O(n_j)$  per update.

### 4.5 Block 4: Update $\tau$ via $\phi = \log \tau$

Since  $\tau > 0$ , we reparametrise  $\phi = \log \tau$  so that the proposal operates on the unconstrained real line.

**Change of variable.** Under the map  $\tau = e^\phi$ , the Jacobian is  $\left|\frac{d\tau}{d\phi}\right| = e^\phi = \tau$ , which contributes an additive  $+\phi$  to the log-density.

**Full conditional for  $\phi = \log \tau$ .** Collecting terms (IV) and (V) from (11) and adding the Jacobian:

$$h(\phi) = -J\phi - \frac{1}{2e^{2\phi}} \sum_{j=1}^J u_j^2 - \frac{e^{2\phi}}{2s_\tau^2} + \phi. \quad (22)$$

**Remark 2** (Why  $\tau$  does not depend on the likelihood). *The likelihood involves  $\mathbf{u}$  but not  $\tau$  directly. Conditional on  $\mathbf{u}$ , the data  $\mathbf{y}$  are independent of  $\tau$ . Data inform  $\tau$  only indirectly through the posterior draws of  $\mathbf{u}$ : large spread in the  $u_j$ 's pulls  $\tau$  upward, while concentrated  $u_j$ 's pull  $\tau$  toward zero. This is a defining feature of the hierarchical structure.*

**Proposal and acceptance.**

$$\phi^* = \phi^{(t)} + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_{\phi, \text{prop}}^2), \quad (23)$$

$$\log R = h(\phi^*) - h(\phi^{(t)}). \quad (24)$$

**Sensitivity variant: Half-Cauchy.** Under  $\tau \sim \text{Half-Cauchy}(0, s_\tau)$ , the last term in (22) is replaced:

$$h_{\text{HC}}(\phi) = -J\phi - \frac{1}{2e^{2\phi}} \sum_{j=1}^J u_j^2 - \log\left(1 + \frac{e^{2\phi}}{s_\tau^2}\right) + \phi. \quad (25)$$

## 4.6 Block 5: Joint Translation Move

**Motivation.** The intercept  $\alpha$  and the random intercepts  $\mathbf{u}$  are *weakly identified* relative to each other: adding a constant  $c$  to  $\alpha$  and subtracting  $c$  from every  $u_j$  leaves  $\eta_i$  unchanged. Blocks 1 and 3 alone explore this “ridge” slowly. A dedicated joint move accelerates mixing dramatically.

**Proposal.**

$$\alpha^* = \alpha + \delta, \quad u_j^* = u_j - \delta \quad \forall j, \quad \delta \sim \mathcal{N}(0, \sigma_{\text{blk}}^2). \quad (26)$$

Because  $\eta_i^* = \alpha^* + \mathbf{x}_i^\top \boldsymbol{\beta} + u_{j[i]}^* = (\alpha + \delta) + \mathbf{x}_i^\top \boldsymbol{\beta} + (u_{j[i]} - \delta) = \eta_i$ , the linear predictor is invariant. Therefore the likelihood cancels exactly in the acceptance ratio, and only the priors matter:

$$\log R = \underbrace{\log p(\alpha^*) - \log p(\alpha)}_{\text{intercept prior}} + \underbrace{\log p(\mathbf{u}^* | \tau) - \log p(\mathbf{u} | \tau)}_{\text{random effect prior}} = -\frac{\alpha^{*2} - \alpha^2}{2\sigma_\alpha^2} - \frac{\sum_j u_j^{*2} - \sum_j u_j^2}{2\tau^2}. \quad (27)$$

**Cost.** This move is  $O(J)$  with *no likelihood evaluation*, so we repeat it  $R = 10$  times per iteration at negligible cost.

## 4.7 Full Iteration Summary

One iteration  $t \rightarrow t + 1$  of the MwG sampler proceeds as:

1. Update  $\alpha$  (Block 1, Section 4.2).
2. For  $k = 1, \dots, K$ : update  $\beta_k$  (Block 2, Section 4.3).

3. For  $j = 1, \dots, J$ : update  $u_j$  (Block 3, Section 4.4).
4. Update  $\phi = \log \tau$  (Block 4, Section 4.5).
5. Repeat  $R = 10$  times: joint translation of  $(\alpha, \mathbf{u})$  (Block 5, Section 4.6).

Each sub-step conditions on the *most recently updated* values of all other parameters. This defines a valid Gibbs scan whose stationary distribution is the posterior (10).

## 5 Posterior Propriety

**Proposition 1.** *Under the model specified in Section 2, the posterior  $p(\alpha, \beta, \mathbf{u}, \tau \mid \mathbf{y}, \mathbf{X})$  is proper (i.e., integrable).*

*Proof sketch.* We verify integrability of each factor:

1. **Likelihood.** For any fixed  $(\alpha, \beta, \mathbf{u})$ , the Bernoulli likelihood is bounded:  $0 \leq p(y_i \mid \eta_i) \leq 1$ , so  $0 \leq p(\mathbf{y} \mid \alpha, \beta, \mathbf{u}) \leq 1$ .
2. **Prior on  $\alpha$ .**  $\alpha \sim \mathcal{N}(0, 25)$  is a proper Gaussian, so  $\int p(\alpha) d\alpha = 1$ .
3. **Prior on  $\beta$ .** Each  $\beta_k \sim \mathcal{N}(0, 6.25)$  independently; the product of  $K$  proper Gaussians is proper.
4. **Conditional prior  $p(\mathbf{u} \mid \tau)$ .** For any fixed  $\tau > 0$ , each  $u_j \sim \mathcal{N}(0, \tau^2)$  is proper.
5. **Hyperprior on  $\tau$ .** The half-normal Half-Normal(0,  $s_\tau^2$ ) is a proper distribution on  $(0, \infty)$ .
6. **Integrability of the joint.** Since the likelihood is bounded above by 1:

$$\int p(\mathbf{y} \mid \cdot) p(\alpha) p(\beta) p(\mathbf{u} \mid \tau) p(\tau) d\alpha d\beta d\mathbf{u} d\tau \leq \underbrace{\int p(\alpha) d\alpha}_{=1} \underbrace{\int p(\beta) d\beta}_{=1} \int p(\tau) \underbrace{\int p(\mathbf{u} \mid \tau) d\mathbf{u}}_{=1} d\tau = 1 < \infty.$$

Since all priors are proper and the likelihood is bounded, the posterior is proper.  $\square$

## 6 Identifiability

**Weak non-identifiability of  $\alpha$  and  $\mathbf{u}$ .** If we define  $\alpha' = \alpha + c$  and  $u'_j = u_j - c$  for any constant  $c$ , the linear predictor  $\eta_i$  is unchanged. This means the likelihood alone cannot separate  $\alpha$  from the mean of  $\mathbf{u}$ . In the Bayesian framework, this is resolved by the priors:

- $p(\alpha) = \mathcal{N}(0, \sigma_\alpha^2)$  anchors  $\alpha$  near zero.
- $p(u_j \mid \tau) = \mathcal{N}(0, \tau^2)$  centers each  $u_j$  at zero.

Together these priors break the translation invariance and make the marginal posteriors of  $\alpha$  and  $\mathbf{u}$  well-defined. Nevertheless, the posterior correlation between  $\alpha$  and  $\bar{u} = J^{-1} \sum u_j$  can be high, leading to slow mixing. The joint translation move (Block 5, Section 4.6) specifically addresses this.



**Fixed effects.** All continuous covariates are standardised (zero mean, unit variance), which prevents collinearity between the intercept and covariate means. Categorical covariates use reference-category coding (Caucasian for race, Circulatory for primary diagnosis), ensuring full-rank design.

## 7 Likelihood-Driven vs. Prior-Driven Behavior

**Fixed effects  $\beta$ .** With  $N = 37,964$  observations and weakly informative priors ( $s_\beta = 2.5$ ), the posteriors of  $\beta$  are overwhelmingly likelihood-driven. The prior contributes negligible shrinkage for well-identified covariates; sensitivity analysis with  $s_\beta = 5$  confirms that posteriors are virtually unchanged.

**Random intercepts  $u_j$ .** The degree of shrinkage depends on group size  $n_j$ :

- **Large groups** (e.g., InternalMedicine,  $n_j = 10,975$ ): the group-specific likelihood dominates, and  $u_j$  is primarily likelihood-driven.
- **Small groups** (e.g., PhysicalMedicine,  $n_j = 288$ ): the prior  $u_j \mid \tau \sim \mathcal{N}(0, \tau^2)$  exerts stronger shrinkage toward zero — this is the *partial pooling* property of hierarchical models.

**Between-group scale  $\tau$ .** The hyperprior  $\tau \sim \text{Half-Normal}(0, 1)$  has moderate influence when  $J$  is small. With  $J = 19$  groups, the posterior of  $\tau$  is primarily determined by the empirical spread of the  $u_j$ 's. The sensitivity analysis (Section 10) confirms this: changing  $s_\tau$  from 1.0 to 2.5 or switching to a half-Cauchy has minimal effect on the posterior of  $\tau$ .

## 8 MCMC Implementation Details

### 8.1 Proposal Tuning

Each proposal standard deviation  $\sigma_{\text{prop}}$  is tuned automatically over 10 rounds of short pilot runs (2,000 iterations each) using a multiplicative adaptation rule: if the acceptance rate exceeds the target (0.44 for scalars, per Roberts & Rosenthal, 2001),  $\sigma_{\text{prop}}$  is multiplied by 1.5; if below target, it is divided by 1.5. Per-coordinate proposal SDs are maintained for each  $\beta_k$  and  $u_j$  to accommodate varying posterior curvatures.

### 8.2 Chain Configuration

- **Number of chains:** 4, with dispersed random initialisations.
- **Iterations per chain:** 30,000 total.
- **Burn-in:** 8,000 (discarded).
- **Thinning:** every 5th sample retained  $\Rightarrow$  4,400 samples per chain, 17,600 pooled.
- **Random seed:** 724 (deterministic reproducibility).

### 8.3 Convergence Assessment

We monitor convergence with two diagnostics applied to every parameter:

**Effective sample size (ESS).** Using Geyer’s initial monotone sequence estimator, ESS measures how many independent draws the autocorrelated chain is worth. We require  $\text{ESS} \geq 100$  for every parameter.

**Split- $\hat{R}$  (Gelman & Rubin, 1992).** Each of the four chains is split in half, producing eight half-chains of length  $n$ . The split- $\hat{R}$  statistic compares between- and within-half-chain variances:

$$\hat{R} = \sqrt{\frac{\hat{V}}{\bar{W}}}, \quad \hat{V} = \frac{n-1}{n} W + \frac{B}{n}, \quad (28)$$

where  $B = \frac{n}{m-1} \sum_{j=1}^m (\bar{\theta}_{j\cdot} - \bar{\theta}_{\cdot\cdot})^2$  is the between-chain variance,  $W$  is the mean within-chain variance, and  $m = 8$  is the number of half-chains. Values close to 1.0 indicate that the chains have mixed well; the conventional threshold is  $\hat{R} < 1.05$ .

#### 8.4 Why Metropolis-within-Gibbs?

The logistic likelihood prevents closed-form full conditionals, ruling out standard Gibbs sampling. Full-dimensional Metropolis–Hastings on the  $(1 + K + J + 1)$ -dimensional parameter space would have a prohibitively low acceptance rate. MwG decomposes the problem into manageable scalar updates, each with its own tuned proposal, yielding robust acceptance rates (target  $\sim 30\text{--}50\%$ ).

## 9 Deviance Information Criterion (DIC)

The DIC provides a Bayesian measure of model fit penalised by complexity:

$$\text{DIC} = \bar{D} + p_D, \quad (29)$$

where

$$D(\boldsymbol{\theta}) = -2 \log p(\mathbf{y} \mid \boldsymbol{\theta}) = -2 \sum_{i=1}^N [y_i \eta_i - \log(1 + e^{\eta_i})], \quad (30)$$

$$\bar{D} = \mathbb{E}_{\boldsymbol{\theta} \mid \mathbf{y}} [D(\boldsymbol{\theta})] \approx \frac{1}{S} \sum_{s=1}^S D(\boldsymbol{\theta}^{(s)}), \quad (31)$$

$$D(\bar{\boldsymbol{\theta}}) = D \text{ evaluated at } \hat{\boldsymbol{\theta}} = \mathbb{E}[\boldsymbol{\theta} \mid \mathbf{y}], \quad (32)$$

$$p_D = \bar{D} - D(\bar{\boldsymbol{\theta}}). \quad (33)$$

Here  $\boldsymbol{\theta} = (\alpha, \boldsymbol{\beta}, \mathbf{u})$  collects all parameters entering the linear predictor, and  $S$  is the number of posterior samples.

$p_D$  is the **effective number of parameters**. For our model with 64 nominal parameters  $(1 + 43 + 19 + 1)$ , the hierarchical prior on  $\mathbf{u}$  induces shrinkage, so we expect  $p_D < 64$ .

## 10 Sensitivity Analysis

We re-run the sampler under three alternative prior configurations:

The goal is to verify that posterior conclusions (especially the magnitude of  $\tau$  and the key fixed effects) are robust to reasonable changes in prior specification.

| Label    | Change                    | $s_\beta$ | $\tau$ prior        |
|----------|---------------------------|-----------|---------------------|
| Baseline | —                         | 2.5       | Half-Normal(0, 1.0) |
| A1       | Wider $\beta$ prior       | 5.0       | Half-Normal(0, 1.0) |
| B1       | Wider $\tau$ prior        | 2.5       | Half-Normal(0, 2.5) |
| B2       | Heavy-tailed $\tau$ prior | 2.5       | Half-Cauchy(0, 2.5) |

Table 2: Prior sensitivity configurations.

## 11 Posterior Predictive Checks

For each of  $S_{\text{rep}} = 200$  posterior draws  $\boldsymbol{\theta}^{(s)} = (\alpha^{(s)}, \boldsymbol{\beta}^{(s)}, \mathbf{u}^{(s)})$ :

1. Compute  $\pi_i^{(s)} = \text{logit}^{-1}(\alpha^{(s)} + \mathbf{x}_i^\top \boldsymbol{\beta}^{(s)} + u_{j[i]}^{(s)})$ .
2. Simulate  $y_i^{\text{rep}(s)} \sim \text{Bernoulli}(\pi_i^{(s)})$  for all  $i$ .
3. Compute test statistics  $T(\mathbf{y}^{\text{rep}(s)})$ :
  - Overall readmission rate:  $T_1 = \bar{y}^{\text{rep}}$ .
  - Specialty rate variability:  $T_2 = \text{SD of group-level rates}$ .

The Bayesian p-value is  $p_B = \Pr(T(\mathbf{y}^{\text{rep}}) \geq T(\mathbf{y}^{\text{obs}}))$ . Values near 0.5 indicate good calibration; values near 0 or 1 indicate systematic model misfit.